

# LEARNHUB

A Mobile Learning & Training Platform Built with Flutter

## WEEK 1 DELIVERABLE — APP PROPOSAL DOCUMENT

|                                     |                                   |                                          |
|-------------------------------------|-----------------------------------|------------------------------------------|
| <b>Project Identity</b><br>LearnHub | <b>Platform</b><br>Flutter / Dart | <b>Target Users</b><br>Learners & Admins |
|-------------------------------------|-----------------------------------|------------------------------------------|

This revised copy standardizes the project name as LearnHub and adds the required Project Objectives section. The GitHub repository details are intentionally marked for completion after the repository is created.

# 1. Purpose of the App

LearnHub is a mobile learning platform that helps organizations deliver structured training programs and helps learners discover, enroll in, and complete courses from their phone. The app solves two problems at once: learners often struggle to find relevant programs and track their own progress, while admins lack a simple mobile-friendly way to publish content, monitor enrollment, and manage learners without relying on a full web dashboard.

The long-term goal is a lightweight, cross-platform (iOS + Android) LMS companion app that complements — rather than replaces — a full web-based admin console, focusing on the on-the-go actions both learners and admins need most.

## 2. Project Objectives

- Provide learners with an easy-to-use platform for discovering and enrolling in training programs.
- Enable learners to track course progress, achievements, and certificates.
- Allow administrators to create, manage, publish, and monitor training programs efficiently.
- Provide a mobile-first learning experience that supports convenient access to training.
- Develop a cross-platform mobile application using Flutter for Android and iOS.
- Support communication and progress monitoring through notifications and learner updates.

## 3. Target Users

### 3.1 Learners

Employees, students, or members of an organization who want to browse available training programs, enroll, track their progress, and view certificates or completion status. They are primarily mobile-first users who expect quick access, minimal friction, and clear progress indicators.

### 3.2 Admins

Program coordinators, HR/L&D staff, or instructors who create and manage training programs, monitor learner enrollment and completion, and communicate updates. Admins need a fast way to check program health and manage content without being tied to a desktop.

## 4. Key Features

### 4.1 Learner Features

- Secure login and profile management (email/password, optional SSO)
- Browse and search a catalog of training programs by category
- View program details: description, duration, modules, prerequisites
- Enroll in a program with one tap
- Track progress through modules with a visual completion indicator
- View and download certificates upon course completion
- Receive push notifications for new programs, deadlines, and reminders
- Personal dashboard showing enrolled, in-progress, and completed programs

### 4.2 Admin Features

- Admin login with role-based access
- Create, edit, and publish training programs and modules
- View list of enrolled learners per program and their progress

- Approve/reject enrollments where required
- Send announcements or reminders to enrolled learners
- Basic analytics: enrollment counts and completion rate per program

## 5. User Journey Examples

### 5.1 Learner Journey — Enrolling in a Program

| Step | Action                               | System Response                                                                                   |
|------|--------------------------------------|---------------------------------------------------------------------------------------------------|
| 1    | Opens the app and logs in            | App authenticates the user and loads the Home screen with recommended and ongoing programs        |
| 2    | Taps "Browse Programs"               | App displays the Program Listing screen with a searchable, filterable catalog                     |
| 3    | Selects a program of interest        | App opens the Program Details screen showing description, duration, and modules                   |
| 4    | Taps "Enroll Now"                    | App confirms enrollment, adds the program to the learner's dashboard, and shows a success message |
| 5    | Opens the program from the dashboard | App loads module content and tracks progress as the learner completes each module                 |
| 6    | Completes all modules                | App marks the program complete and generates a downloadable certificate                           |

### 5.2 Admin Journey — Publishing a New Program

| Step | Action                              | System Response                                                                     |
|------|-------------------------------------|-------------------------------------------------------------------------------------|
| 1    | Logs in with an admin account       | App authenticates and routes the user to the Admin Home/Dashboard                   |
| 2    | Taps "Create Program"               | App opens a form to enter program title, description, category, and duration        |
| 3    | Adds modules and content            | App saves each module in draft state under the new program                          |
| 4    | Taps "Publish"                      | App validates required fields and makes the program visible in the learner catalog  |
| 5    | Opens "Enrollments" for the program | App shows a list of enrolled learners with individual progress status               |
| 6    | Sends a reminder announcement       | App pushes a notification to all enrolled learners who have not started the program |

## 6. Tech Stack & Navigation

The app is built in Flutter (Dart 3.x, Material 3) to support both iOS and Android from a single codebase. Routing uses GoRouter, with Provider for state management in Week 1–2 (Riverpod planned for later weeks). The backend will integrate Firebase Authentication and Firestore starting Week 3, with SharedPreferences used for local/offline progress caching.

Navigation branches by role immediately after login: learners land on a Home Dashboard with Program Listing, My Progress, and Profile; admins land on an Admin Dashboard with Create Program, Manage Learners, Assignments & Quizzes, Reports & Analytics, and Settings.

### Navigation Flow

Login Screen

```
■■■■ [Learner] → Home Dashboard
■   ■■■■ Program Listing → Program Detail → Enroll
■   ■■■■ My Progress
■   ■■■■ Profile
■■■■ [Admin] → Admin Dashboard
    ■■■■ Create Program
    ■■■■ Manage Learners
    ■■■■ Assignments & Quizzes
    ■■■■ Reports & Analytics
    ■■■■ Settings
```

## 7. Wireframe Designs

UI/UX wireframe designs for the four core screens — Login, Home Dashboard, Program Listing, and Program Details — are included separately as screenshots. They outline the basic layout of navigation bars, buttons, text areas, forms, and screen transitions for the learner flow.

**Core screens: 01 Login Screen   02 Home Dashboard   03 Program Listing   04 Program Details**

## 8. GitHub Repository Setup

The GitHub repository will be created as part of the Week 1 technical setup. The final repository name will be **learnhub\_flutter**. The repository link and evidence screenshots should be added after the GitHub setup is completed.

**Repository:** To be added after GitHub setup

### 8.1 Repository Structure

```
learnhub_flutter/
■■■■ lib/
■   ■■■■ main.dart
■   ■■■■ screens/
■   ■   ■■■■ login_screen.dart
■   ■   ■■■■ home_screen.dart
■   ■   ■■■■ program_listing_screen.dart
■   ■   ■■■■ program_detail_screen.dart
■   ■   ■■■■ admin_dashboard_screen.dart
■   ■■■■ widgets/
■   ■■■■ models/
■   ■   ■■■■ user.dart
■   ■   ■■■■ program.dart
■   ■■■■ services/
■       ■■■■ auth_service.dart
■       ■■■■ program_service.dart
■■■■ assets/
■■■■ docs/
■■■■ android/
■■■■ web/
■■■■ test/
■■■■ pubspec.yaml
```

## 8.2 Version Control Workflow

The repository will use GitHub for version control. Week 1 evidence should include at least one commit/push, the initial Flutter project structure, and a README describing the project vision, objectives, target users, core features, navigation flow, project structure, technology stack, and roadmap.

## 8.3 Roadmap

| Week   | Paragraph( 'caseSensitive': 1 'encoding': 'utf8' 'text': 'Deliverables' 'frags': [ParaFrag(__tag__='para', bold=1, fontName='Helvetica-Bold', fontSize=8.3, greek=0, italic=0, link=[], rise=0, text='Deliverables', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0 ) #Paragraph |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1 | Project setup, navigation, screen stubs, wireframes, proposal document                                                                                                                                                                                                                                           |
| Week 2 | Functional UI screens, form validation, static data rendering                                                                                                                                                                                                                                                    |
| Week 3 | Firebase integration (Authentication + Firestore), real data                                                                                                                                                                                                                                                     |
| Week 4 | Progress tracking, quiz engine, certificate generation                                                                                                                                                                                                                                                           |
| Week 5 | Testing, polish, Play Store / App Store submission preparation                                                                                                                                                                                                                                                   |

**Submission note: The final Week 1 submission can combine the App Proposal, wireframe screenshots, and GitHub repository link/evidence in one PDF, as required by the course brief.**